Materials theory and modelling

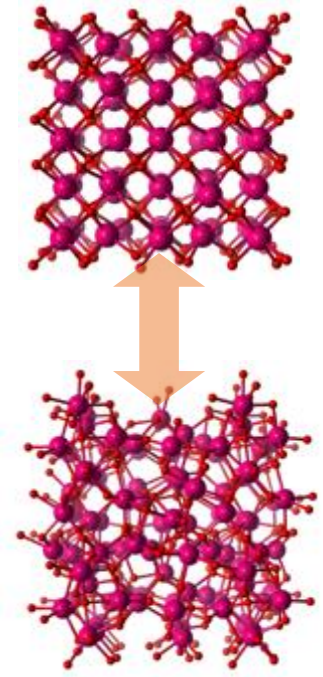
Can we from the potential energy between interacting objects, electrostatics and Newton's equations of motion design, predict and understand the atomic structure of materials to advance technological progress?

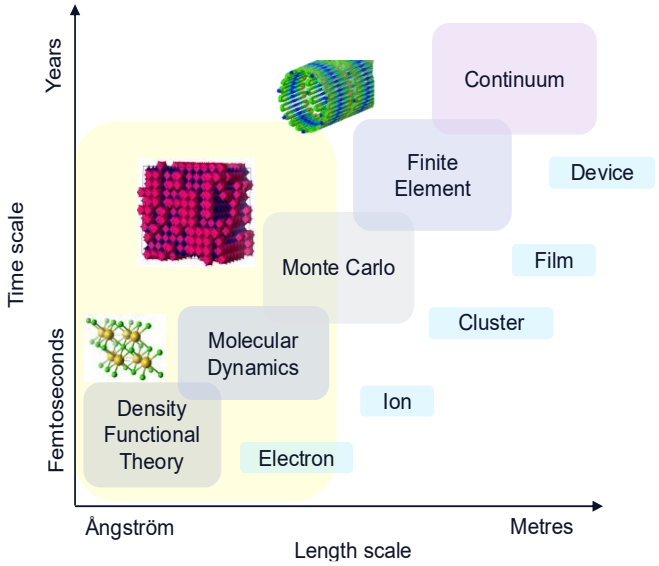
Proposed projects:

1 – How does the morphology of thin films affect electronic properties?

2 – How can we optimise ionic diffusion through lattice manipulation?

3 – How does tensile and compressive stress affect the properties of binary and tertiary systems?





Courses in condensed matter physics, Linux literacy, and programming skills are assets.

We also welcome project ideas on coatings, memory, fuel cells, 2D materials, batteries, catalysis and surface interactions.

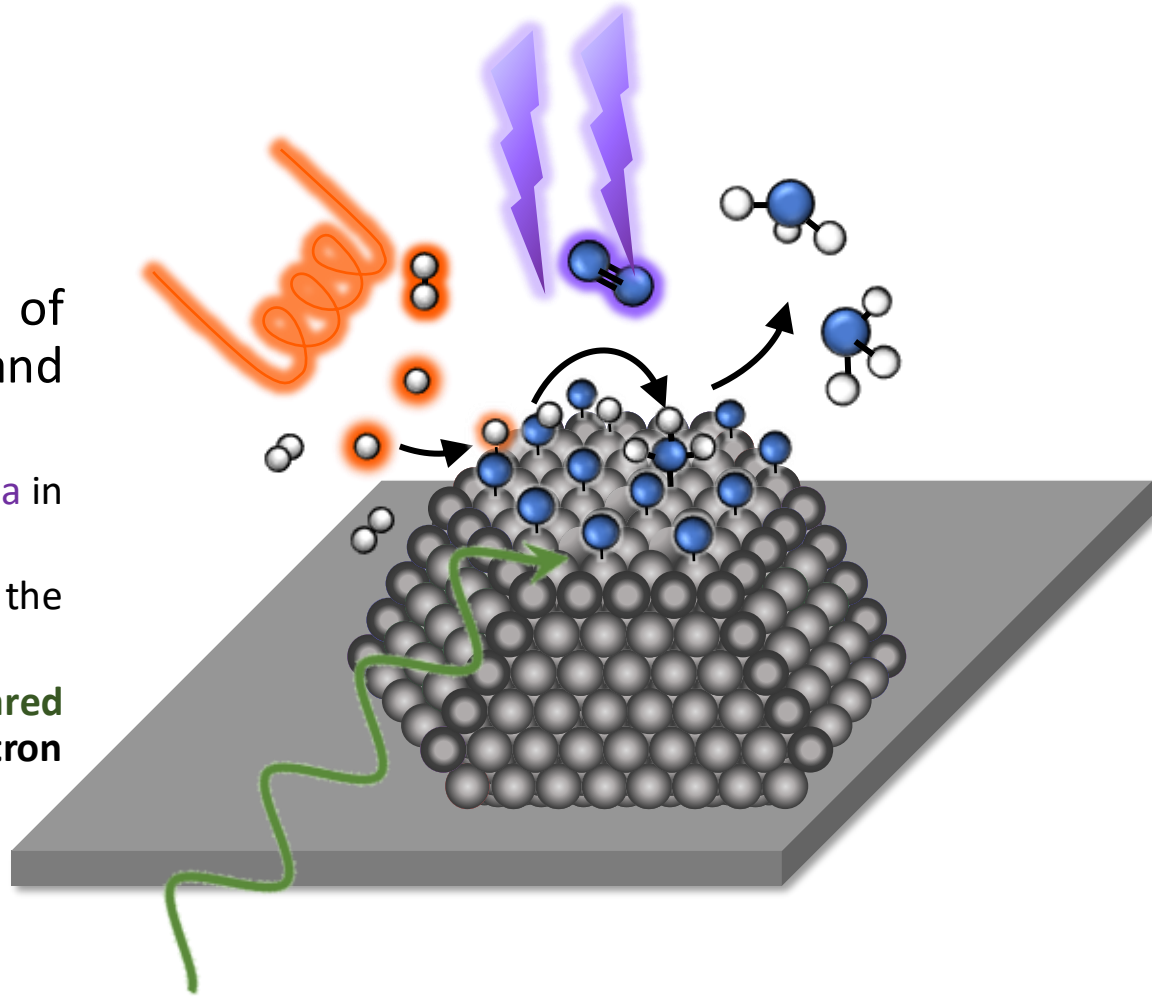
Contact: Emilia Olsson at olsson@arcnl.nl . Website: https://arcnl.nl/research-groups/materials-theory-and-modeling

Resolving mechanisms of plasma catalysis using in situ IR spectroscopy

Motivation:
Plasma activation of molecules enhances catalytic reactions at surfaces. The surface reaction mechanisms and the active species are still unknown.

In this project, you will:

- Determine which surface species contribute to reactions of catalytically active metal surfaces in plasma-activated N_2 and H_2.
  - Grow nitride films on single-crystalline metal surfaces, using N_2 plasma in ultra-high vacuum setups
  - Reduce them using atomic hydrogen, during spectroscopy changes in the spectroscopic signature.
  - Characterize their molecular bonds and structure using **infrared reflection absorption spectroscopy (IRRAS)** and **low-energy electron diffraction** (LEED)
  - Monitor products in the gas phase by **mass spectrometry**.



ARCNL

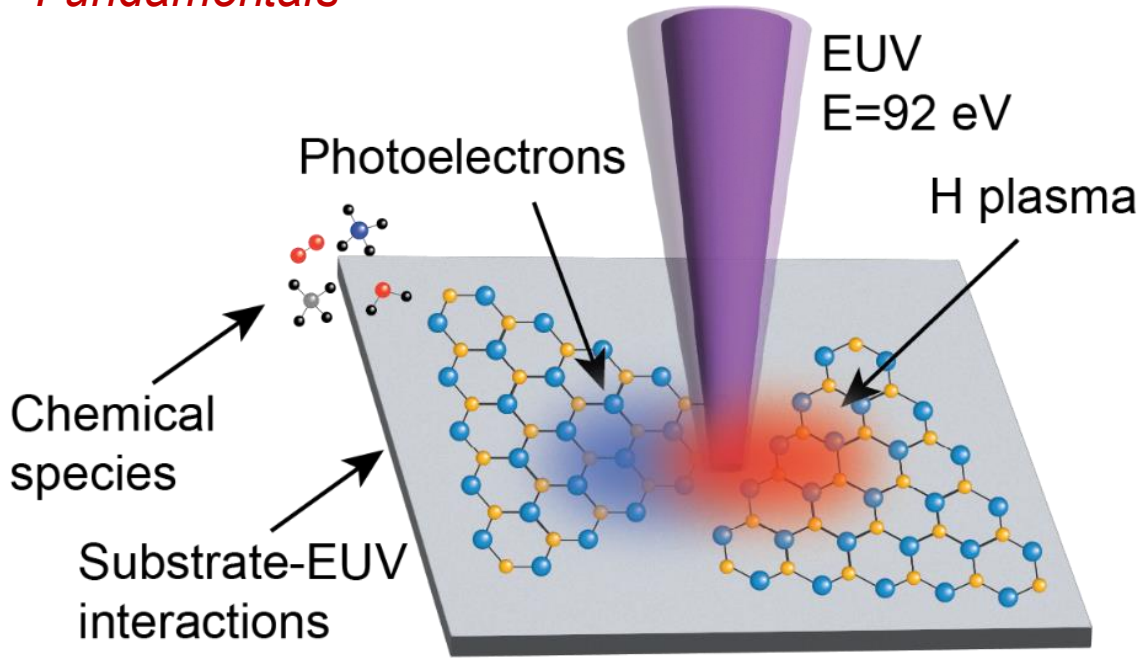
Sabine Auras,
Roland Bliem

Interested? Contact bliem@arcnl.nl

Patterning 2D Semiconductors with EUV light

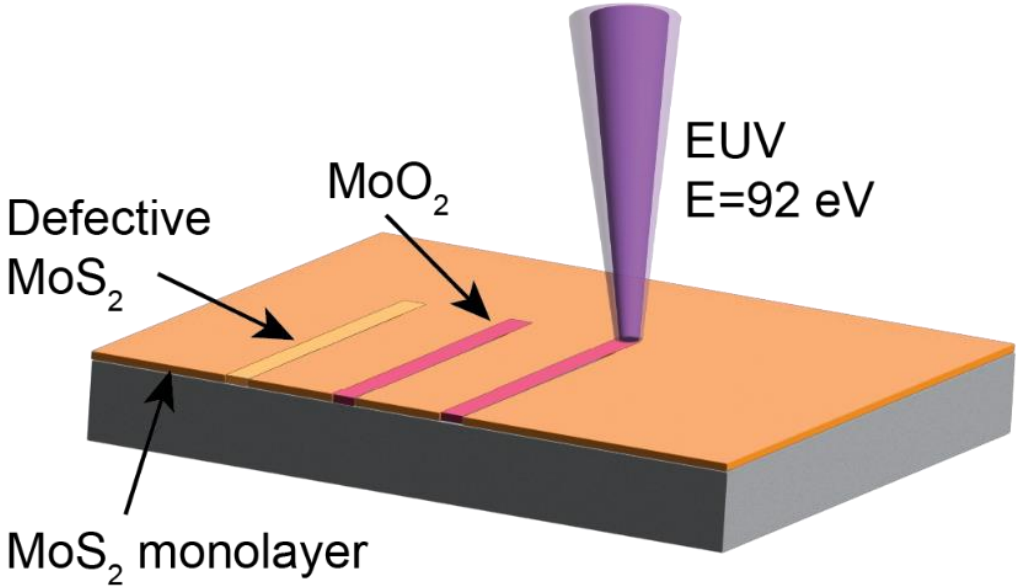
We study fundamental interactions of 2D semiconductors with extreme ultraviolet (EUV) lithography environments to design new patterning approaches for next-generation chips

Fundamentals



EUV lithography parameter space

Patterning approaches



Selective EUV chemical transformations

ARCNL

University of Amsterdam

Contact

Dr. Reynolds Dziobek-Garrett

Dr. Roland Bliem (bliem@arcnl.nl)

Collaboration: Dr. Jorik van de Groep

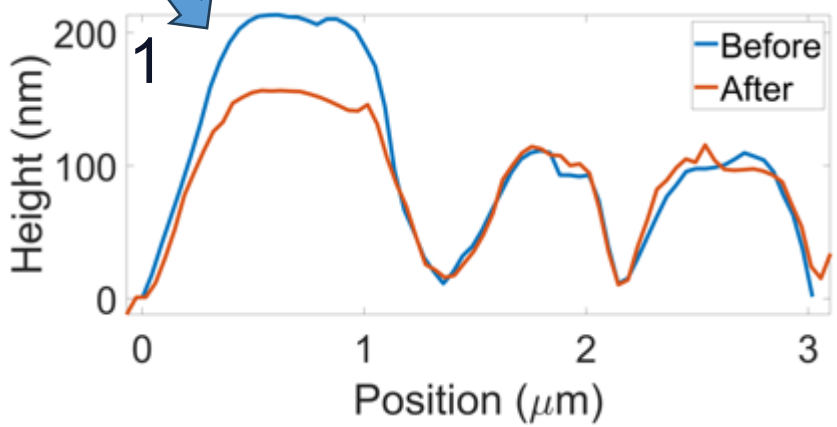
ARCNL

Diamonds are forever?

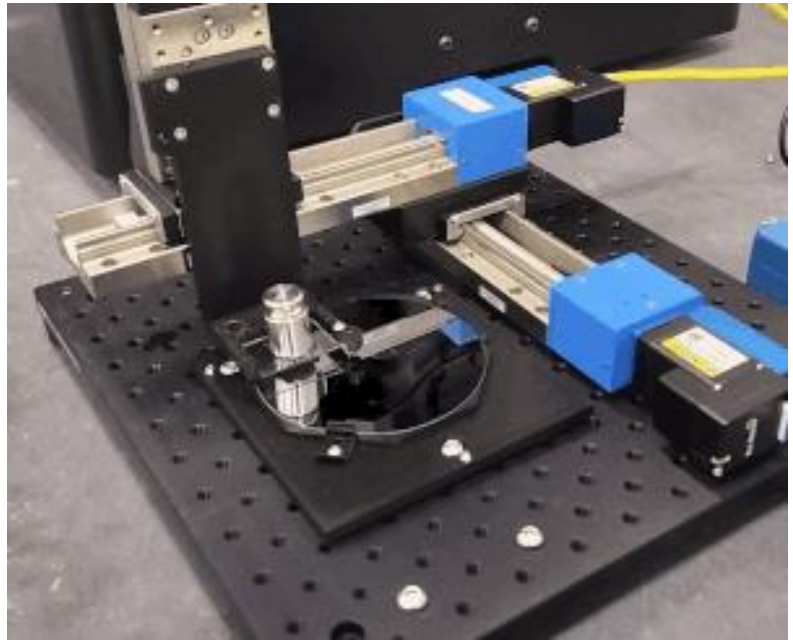
No! Even Diamond wears, when you subject it to $23 \cdot 10^9$ Pa (!) pressure.

What happens at lower contact pressures? You will help us find out in this BSc project.

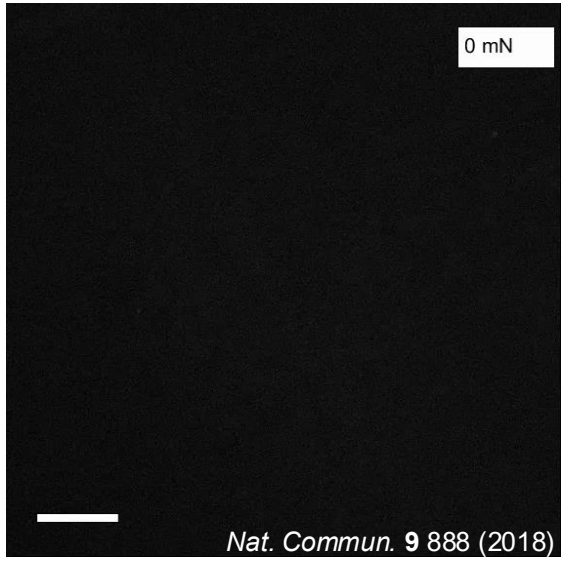
Atomic Force Microscopy



Wear Experiments



Contact Calculations



Bart Weber

University of Amsterdam

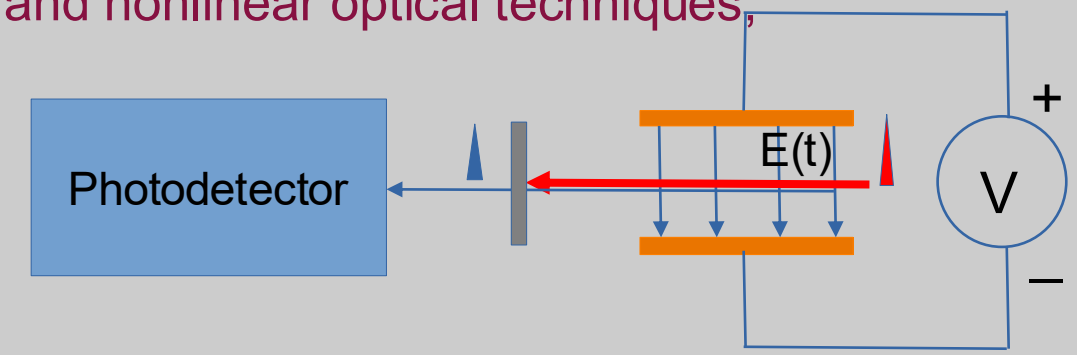
Light/Matter interaction group BSc project:
Measuring electric fields in air

- Under many circumstances, such as airflows, friction, or just (un)intentional charging of materials, electric fields are created that extend into the air

- It is important to measure these electric fields but conventional probes tend to perturb them, complicating the analysis

- **In this Bsc you will use femtosecond laser pulses** to determine the strength and direction of electric fields in air, using a nonlinear-optical technique called Electric Field Induced Second Harmonic Generation (EFISH)

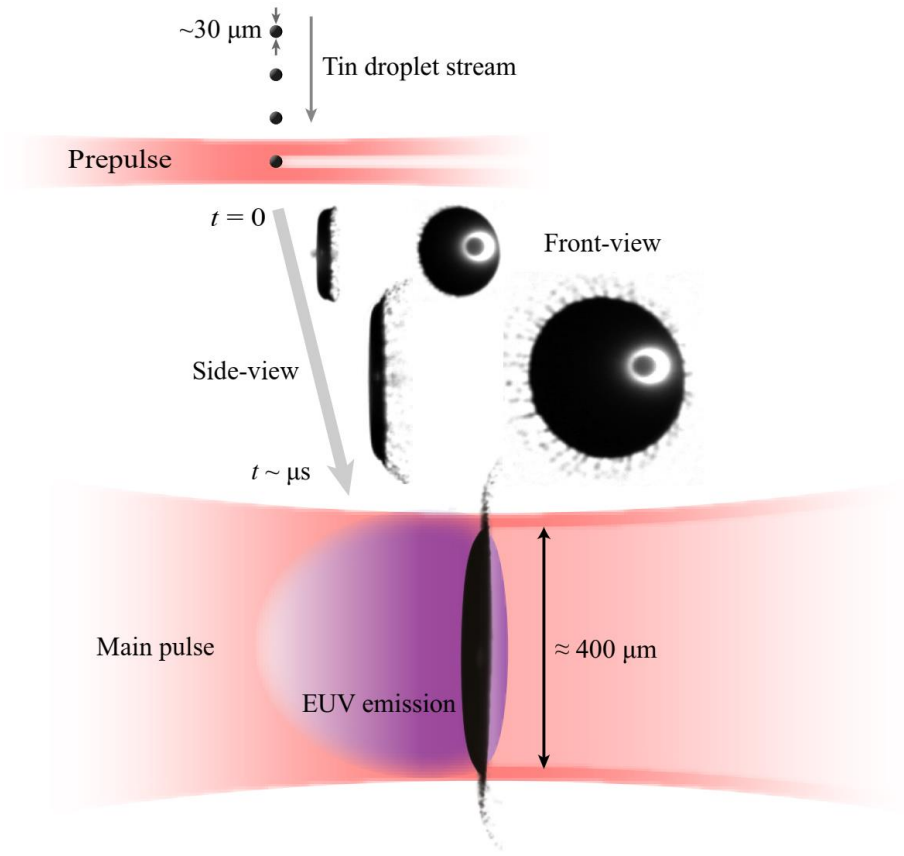
- **What you will learn:**
- Working with state-of-the-art high-power femtosecond lasers,
- Working with photodetectors, linear and nonlinear optical techniques,
- data analysis,….



Contact: Paul Planken planken@arcnl.nl

ARCNL Source Research: physics questions

Extreme ultraviolet (EUV) lithography enables producing the most advanced computer chips. At the source of this technology lies a hot and dense laser-produced tin plasma. This plasma produces intense bursts of EUV radiation. The ever-increasing hunger for advanced technologies, more powerful computer chips, asks for the further development of high-power EUV sources.
At **ARCNL** we wonder: what will the more powerful and more energy-efficient EUV sources of the future look like?



1. **Understand exploding tin microdroplets**
   - What determines deformation and fragmentation?
2. **Key insights to enable source predictive modeling**
   - What emits that EUV light?
3. **Push the fundamental limits of the conversion efficiency**
   - What sets the fundamental limit?
4. **Control expansion dynamics of laser-produced plasma**
   - What is the cause of the ion energy distribution, how can it be controlled, and what does it cause?

Contact: Oscar Versolato at versolato@arcnl.nl. Website: https://arcnl.nl/research-groups/euv-plasma-processes

EUV Plasma Processes group

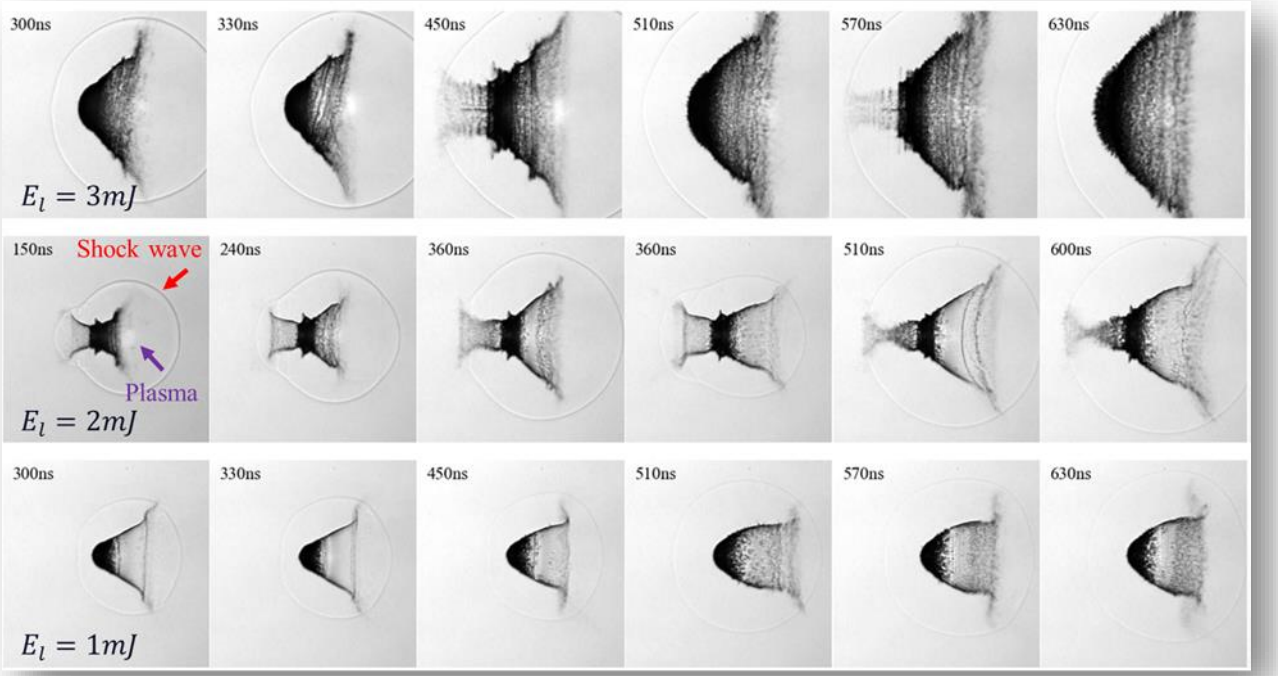
We explode water droplets with a laser. This is relevant to understanding target shaping for generating EUV light for modern nanolithography (we can explain). But… how can we explain what happens here?

Project A:
Laser-exploding water droplets: experiments

Help us **perform experiments** using our laser-droplet setup at ARCNL to find out!

Project B:
Laser-exploding water droplets: modeling

Help us **perform fluid-dynamic simulations** using Basilisk to find out!



Help us answer these questions (and many others)! Interested in these projects or our research in general?

Contact: Oscar Versolato at versolato@arcnl.nl. Website: https://arcnl.nl/research-groups/euv-plasma-processes

Ultrashort pulse characterisation

Research group: High Harmonic Generation & EUV Science at Advanced Research Center for Nanolithography (ARCNL).

Contacts PI: Dr. Peter Kraus | | P.Kraus@arcnl.nl | +31208517100

Postdoc: Dr. Anchit Srivastava | A.Srivastava@arcnl.nl| +31208517100

Description

Ultrashort laser pulses are a cornerstone of modern science and technology. With durations in the femtosecond to attosecond range (10^{-15} –10^{-18} s), they provide the temporal resolution needed to observe and control the fastest processes in nature. These pulses enable real-time studies of electron motion, chemical reactions, and energy transfer on their fundamental timescales, opening the door to breakthroughs in physics, chemistry, materials science, and beyond. Before such pulses can be used in experiments, however, their properties must be precisely characterized. Only by knowing the exact temporal and spectral shape of the pulse beforehand can we determine how it has been modified by its interaction with a sample. Pulse characterization is therefore not just a diagnostic step — it is essential for extracting meaningful physical insight from ultrafast experiments.

In this project, you will develop a pulse characterization technique known as Frequency-Resolved Optical Gating (FROG). Further you'll implement a retrieval algorithm to reconstruct ultrashort optical pulses from measured spectrograms. You will explore the physics of nonlinear pulse characterization and simulate FROG traces numerically. In addition, you will implement and optimize an iterative phase-retrieval algorithm, improving its convergence and testing its robustness against noise and experimental imperfections.

You'll gain a solid foundation in Fourier optics and ultrafast pulse physics while developing practical skills in numerical optimization, signal processing, phase retrieval, and scientific programming (Python).

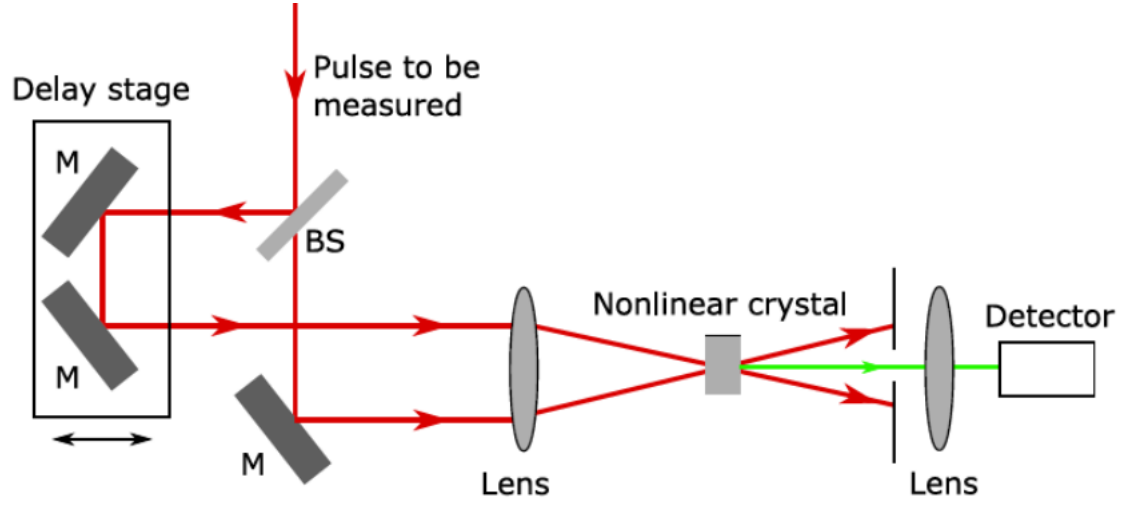


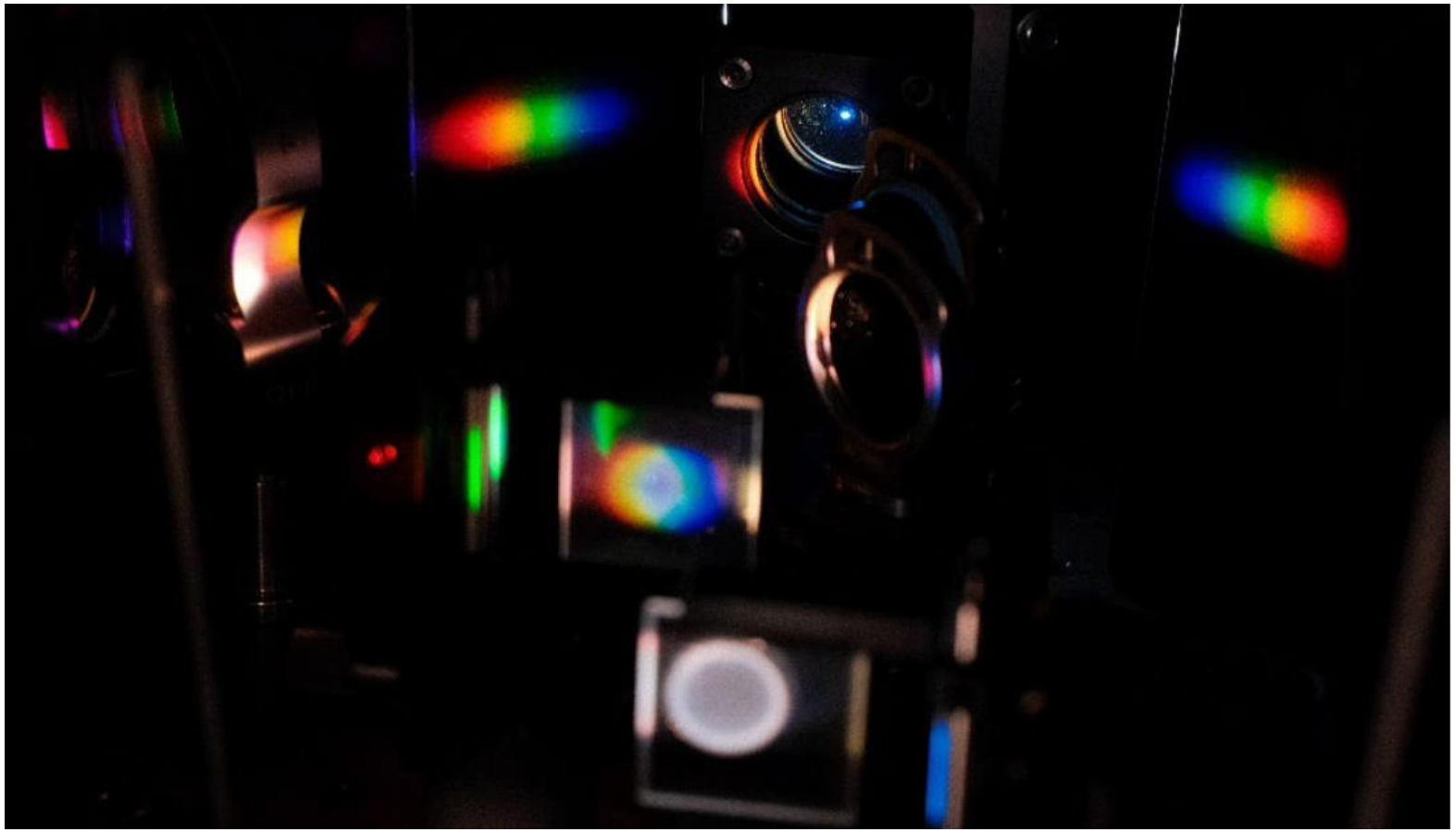
Figure1: Schematic of Autocorrelation and frequency-resolved optical gating measurement setup. A diagram illustrating a non-collinear setup for second-harmonic generation or frequency-resolved optical gating measurement. In this configuration, a photodiode serves as the detector for autocorrelation measurements, while a spectrometer is employed for FROG measurements. BS: beamsplitter; M: silver mirror.

Probing doping and defects in semiconductor materials using high-harmonic generation

Research group: High Harmonic Generation & EUV Science at Advanced Research Center for Nanolithography (ARCNL).

Peter Kraus, Advanced research center for Nanolithography (ARCNL), p.kraus@arcnl.nl

Pieter van Essen, Advanced research center for Nanolithography (ARCNL), p.essen@arcnl.nl



Probing doping and defects in semiconductor materials using high-harmonic generation

Research group: High Harmonic Generation & EUV Science at Advanced Research Center for Nanolithography (ARCNL).

Peter Kraus, Advanced research center for Nanolithography (ARCNL), p.kraus@arcnl.nl

Pieter van Essen, Advanced research center for Nanolithography (ARCNL), p.essen@arcnl.nl

Description

Doped semiconductor materials play a key role in electronic devices and integrated circuits. The continuous miniaturization of the active components on chips have further increased the necessity of highly controlled material quality. For high quality devices consistent quality control is required, which must be very sensitive, high-resolution, very fast and non-destructive. While the current standard for inspection is electron microscopy, optical techniques have the potential to greatly improve the detection speed while vastly reducing damaging capabilities. But to reap these benefits development is required to overcome limitations of resolution and sensitivity.

While doping and defects might greatly impact the electronic properties of a materials these changes are often not reflected in their optical properties. This results in little to no contrast in conventional optical microscopes, limiting the possibility for all-optical inspection. By going beyond conventional microscopy to non-linear techniques where new wavelengths are generated within the sample, this limitation can be overcome. Here we will employ high-harmonic generation for this purpose.

The harmonic process is greatly impacted by the electronic properties of the material making it highly suitable for the detection of material doping and defects. While great sensitivity to doping and defects is to be expected, how these will exactly affect the high-harmonic process is an open question.

In this project the aim is to better understand how material doping and defects can impact the harmonic generation process. The project will evolve around optical experiments where the high- harmonic generation from a plethora of differently doped samples is measured and studied. 2 This is a project in the HHG & EUV science group of Peter Kraus at the Advanced Research Center for Nanolithography (ARCNL). This project is for if you have an interest in optics, optical experiments, and condensed matter. If you want to know more about the fundamental physics and context behind this project, we recommend looking through Ref. [1].

Tasks or milestones - Work (independently) on the experimental setup designed for measuring harmonic emission from solids. - Measure the (time-resolved) harmonics from various doped samples. - Analyze and evaluate highly dimensional measurement data. - Improve the understanding of defect/doping on the high-harmonic generation process - Discuss findings at weekly (group) meetings. - Present the final thesis results. - Write a final report.

References

[1] Pieter J. van Essen, Zhonghui Nie, Brian de Keijzer, and Peter M. Kraus, Toward Complete All- Optical Intensity Modulation of High-Harmonic Generation from Solids, ACS Photonics 2024 11 (5), 1832-1843, 10.1021/acsphotonics.4c00156

Measuring ultrashort light pulses generated by nonlinear optics

Research group: High Harmonic Generation & EUV Science at Advanced Research Center for Nanolithography (ARCNL).

Peter Kraus, Advanced research center for Nanolithography (ARCNL), p.kraus@arcnl.nl

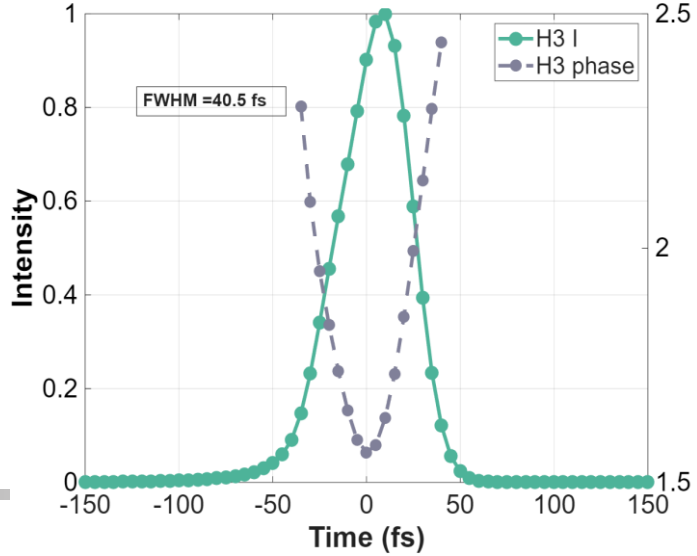
Daily supervisor: Torben Ham, PhD student ARCNL; t.ham@arcnl.nl

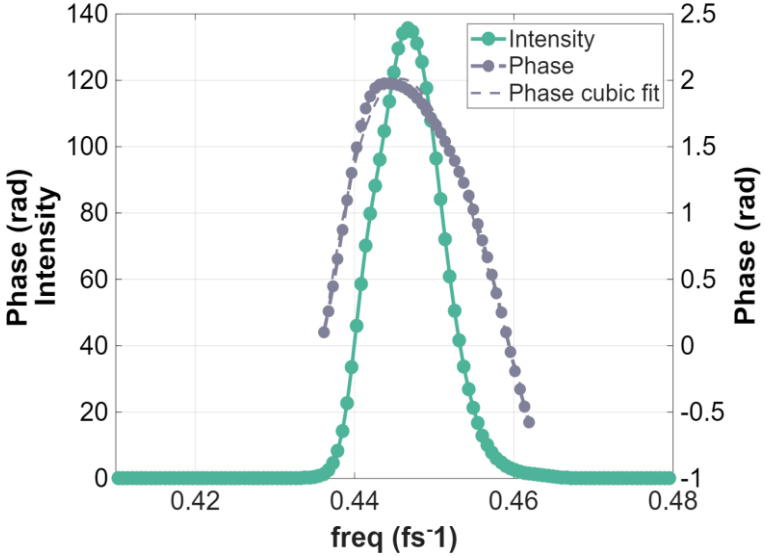
Description

The harmonic emission that results from focusing ultrashort laser pulses in a medium can carry sensitive information about modifications of the electronic structure of the medium on femtosecond timescales. Measuring the full field of the harmonic pulses, however, is nontrivial as the phase of the electric field needs to be measured. Recently, we have developed a setup in which both the amplitude and the phase of the laser pulses can be retrieved. This setup needs to be very well characterized by measuring harmonic emission from different materials and with different generation pulses, with the end goal having a solid understanding of how the setup can be used for interpreting measurements on strongly correlated materials with an insulator-to-metal transition. The student will work on improving the setup, doing experiments for a good characterization, and possibly adding a pump-probe scheme.

Tasks or milestones

- Improve and characterize the setup. - Measure nonlinear optical pulses generated from different materials: both regular insulators and more exotic materials such as correlated oxides. - Analyze data and use the results to improve on the signal detection - Write a final report on the work and present the work during a group meeting





ARCNL

AMSTERDAM SCIENCE PARK
City Center
Ooster Ringdijk (cycle route)
Diemen - IJburg
Matrix Innovation Center ARCNL
106
110
Digital Realty
120
P1A
P1T
P1B
NLeSC
140
SURFsara
Science Park
Science Park
Science Park
402
404
400
Matrix Innovation Center
P4
406
408
ILCA
Coffee Virus
410
Science & Business organisation
Universum
Oerknal
306
104
AMOLF
102
129
131
CWI
125
123
135
Interxion
121
Science Park Aqua
Bicycle repair shop
105
Nikhef
103
Residential housing
Carolina MacGillavryLaan
Student accommodation
UvA
107
P1E
P3
Science Park
111
AUC
113
Science Park Aer
Science Park
Spar
Maslow
201
Polder
UvA Faculty of Science
904
IXA
Science Park
608
Startup Village
ACE Incubator
Coffee Virus
610
650
Equinix
Datatower AM4
Science Park Terra
Kruislaan
Anna's Tuin en Ruigte
S113
Ringweg A10
Carolina MacGill
Spark Village
P7